

Psychological Mechanisms and Intervention Strategies for College Students' Short-Video Addiction in Algorithmic Recommendation Scenarios

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Abstract

With the rapid proliferation of short-video platforms, short-video addiction among college students has become increasingly prominent, posing a significant challenge to both the quality of talent cultivation in higher education and the psychological and physical well-being of students. This paper systematically analyzes the psychological mechanisms underlying college students' short-video addiction from the perspective of algorithmic recommendation technologies. At the neural level, micro-duration stimuli optimized by algorithms trigger dopamine release through variable reward schedules, thereby generating conditioned usage impulses. At the psychological level, drawing upon the I-PACE (Interaction of Person-Affect-Cognition-Execution) model, the interaction between individual factors—such as time orientation, dual-system self-control, loneliness, and social support—and cognitive-emotional responses jointly drives the onset and maintenance of addictive behaviors. At the environmental level, information cocoon effects and peer social pressure further exacerbate addiction. On this basis, this paper proposes a multi-dimensional tiered intervention strategy: at the individual level, focusing on self-control training and future-oriented cognitive reappraisal; at the platform level, promoting algorithmic transparency and embedding anti-addiction design features; at the institutional level, developing mental health education and media literacy curricula; and at the societal level, refining algorithmic governance regulations and collaborative intervention networks. This study aims to provide a theoretical foundation and practical pathways for the scientific prevention and treatment of short-video addiction among college students.

Keywords

Algorithmic Recommendation; Short-Video Addiction; College Students; Psychological Mechanism; Intervention Strategies

1. Introduction

Short-video applications have become the predominant form of digital

entertainment and social media among contemporary college students. As of December 2025, the number of online video users in China reached 1.093 billion, an increase of 22.43 million compared to December 2024, accounting for 97.1% of the total netizen population. Among them, short-video users numbered 1.074 billion, representing 95.4% of all netizens (China Internet Network Information Center, *57th Statistical Report on China's Internet Development*). College students constitute a primary consumer group of short-video content (Fu & Cheng, 2026). Moderate usage facilitates information acquisition and emotional regulation; however, excessive use engenders pronounced negative consequences, including diminished academic concentration, weakened social skills, heightened anxiety and depression, and the eventual development of behavioral addiction (Li & Bao, 2026; Shang et al., 2025).

Unlike traditional social media, short-video platforms rely on neural network-based recommendation systems to continuously optimize content matching. Coupled with micro-duration stimulus designs, these platforms trigger dopamine release via variable reward schedules, propelling users into an "auto-pilot mode" of persistent scrolling (Mao & Liao, 2025). This algorithm-driven "addictive design" has attracted international regulatory scrutiny. In 2026, the European Union, under the Digital Services Act (DSA), launched an investigation into TikTok, citing that its algorithmic recommendation mechanisms and interface features—including infinite scrolling, auto-play, and push notifications—constitute systematic manipulative design that impairs users' capacity to make free and informed decisions.

College students are situated in a critical developmental phase characterized by unstable self-identity and incompletely matured executive functions. Their heightened psychological sensitivity and variable self-control capacities render them particularly susceptible to algorithmic recommendation systems. Therefore, systematically elucidating the psychological mechanisms of short-video addiction among college students within algorithmic recommendation contexts, and devising precise intervention strategies accordingly, carries substantial theoretical and practical significance.

2. Psychological Mechanisms of Short-Video Addiction in Algorithmic Recommendation Scenarios

2.1. Etiological Analysis of Short-Video Addiction in Algorithmic Recommendation Scenarios

The etiological factors of short-video addiction in algorithmic recommendation scenarios can be delineated across three dimensions: technological, psychological, and individual.

2.1.1. Technological Dimension: The Algorithmic "Addiction Loop"

The core driving force behind short-video platforms is the personalized algorithmic recommendation system. By aggregating user behavioral data—including browsing

history, likes, and comments—and employing content-based filtering and collaborative filtering techniques, recommendation algorithms precisely deliver content to users' mobile devices. This establishes a closed-loop cycle of "personalized → user need satisfaction → behavioral data accumulation → algorithm optimization → re-push." Within this loop, continuous algorithmic refinement increases the precision of information delivery, fostering user dependence on, and diminished self-control over, short-video consumption. Research indicates that algorithmic recommendations enhance users' immersion and dependence through highly personalized content, while information cocoons restrict access to diverse information, thereby exacerbating cognitive limitations and behavioral dependency.

2.1.2. Psychological and Neural Mechanisms: Dopamine-Driven "Reward Circuitry"

Viewing personalized short-video content continuously stimulates the ventral tegmental area (VTA)—a neural circuit central to the generation of pleasure and motivation. Sustained activation of this region induces craving and addictive states. From a neuroscience perspective, dopamine is not predominantly released upon obtaining a reward; rather, it surges during the *anticipation* of an impending reward (i.e., the moment of expectancy before scrolling to the next video), entrapping users in a cycle of "pursuit → transient satisfaction → further pursuit." Prolonged high-frequency stimulation may further induce the expression of Δ FosB (a "addiction protein"), altering gene expression within reward circuitry.

Short videos also capitalize on instant gratification mechanisms, often referred to as the "sunflower seed effect." Their 15-second vertical, high-stimulation format, combined with dynamic visuals and music, continuously captures attention, disrupts the brain's default mode network, and compromises self-regulatory capacity.

2.1.3. Individual and Content Dimensions: Low Operational Cost and User Characteristics

Short videos are extremely brief (15–60 seconds), and users can infinitely extend viewing sessions with a simple swipe, entailing minimal operational cost and facilitating effortless, prolonged engagement. Data indicate that Douyin (TikTok China) once achieved 400 million daily active users, with an average daily usage frequency exceeding 21 sessions and a duration nearing 60 minutes; 41.3% of users engage with the platform multiple times daily, and 56.9% use it for over 60 minutes per day. Among adolescents, whose psychological development remains immature and self-control abilities are comparatively weaker, susceptibility to the instant feedback and highly stimulating content of short videos is particularly pronounced, predisposing them to addiction.

2.2. Psychological Mechanisms of Short-Video Addiction in Algorithmic Recommendation Scenarios

2.2.1. Neuro-Psychological Basis: Dopaminergic Reward Pathways and Conditioned Reflexes

Addiction can be conceptualized as a pathological learning process whereby behavior "hijacks" the brain's reward system. Nearly all addictive behaviors share a common neurobiological effect: a sharp, supraphysiological surge in dopamine levels within the mesolimbic pathway. The content-switching mechanism of short-video platforms is structurally isomorphic to slot-machine variable rewards—each downward swipe carries the potential for encountering more engaging content, and this unpredictable reward probability maximally activates the dopaminergic system. Chronic, repeated stimulation leads to downregulation of dopamine receptors and elevation of reward thresholds, thereby diminishing the hedonic value of natural rewards (e.g., learning, social interaction, and physical exercise), forming a vicious cycle of "increased usage breeds dissatisfaction, and dissatisfaction drives further usage."

At the behavioral level, algorithmic recommendation systems are designed to establish sustained, automated usage habits, generating conditioned usage impulses at the neural level that progressively undermine the user's capacity to govern usage behavior through rational decision-making. Framed within Sunstein's theory of manipulative choice architecture, algorithmic recommendation systems bypass users' deliberative decision-making processes (System 2) and predominantly engage intuition- and impulse-driven automatic responses (System 1), thereby inducing usage decisions made without adequate deliberation.

2.2.2. Multi-Factor Interaction Mechanisms from the I-PACE Model Perspective

The I-PACE (Interaction of Person-Affect-Cognition-Execution) model, proposed by Brand et al. (2019), provides an integrative theoretical framework for understanding short-video addiction in algorithmic recommendation scenarios. This model posits that addictive behaviors emerge from the reciprocal interaction among individual characteristics, affective-cognitive responses, executive functions, and situational factors.

Temporal Orientation and Dual-System Self-Control. A large-sample study (N = 2,239) grounded in the I-PACE framework revealed that past- and present-oriented temporal foci are significantly positively correlated with short-video addiction, whereas future-oriented temporal focus is negatively correlated. Inhibitory self-control mediated the relationship between past/present orientation and addiction—suggesting that individuals excessively focused on the past or present tend to employ inhibitory strategies against impulses; however, inhibition itself may consume cognitive resources, resulting in a "resistance backfire effect" (the harder one resists, the deeper one becomes entrenched). Conversely, initiatory self-control mediated the negative predictive effect of future orientation on addiction—individuals with clear future goals are capable of actively initiating beneficial behaviors to replace habitual scrolling routines. Person-centered analyses

further classified college students' short-video addiction into four categories: non-addiction (12.68%), low-risk addiction (34.21%), moderate-risk addiction (42.20%), and high-risk addiction (10.89%), indicating pronounced population heterogeneity that necessitates differentiated intervention strategies (Liu et al., 2025).

Chain Mediation of Loneliness, Social Support, and Physical Activity. Questionnaire-based research involving 388 college students demonstrated that loneliness significantly and positively predicts short-video addiction, with each unit increase in loneliness corresponding to a significant rise in addiction propensity (Pan, 2005). Social support and physical activity independently mediated the relationship between loneliness and short-video addiction, and they formed a chain mediation pathway: loneliness → diminished perceived social support → reduced physical activity → increased short-video addiction. This finding elucidates the social substitution hypothesis mechanism—while short videos may temporarily alleviate loneliness, they ultimately exacerbate the reciprocal reinforcement of loneliness and addiction by displacing time allocated for genuine social interaction and physical exercise.

The Double-Edged Sword Effect of Cognitive-Affective Dynamics. Recent research (N = 1,274), also framed within the I-PACE model, found that the efficacy of self-regulation strategies is conditional. Among individuals with low inhibitory control, greater utilization of self-regulation strategies paradoxically correlated with higher addiction severity (a "backfire effect"), suggesting that simply instructing low-self-control individuals to "exercise restraint" may be counterproductive. Conversely, for those with high inhibitory control, self-regulation strategies were largely redundant. Furthermore, the addiction pathways of addicted versus non-addicted individuals were qualitatively distinct: addicted individuals' behavior was predominantly driven by the pursuit of positive affect, whereas non-addicted individuals' critical defense was self-regulatory capacity—offering a precise window for early intervention (Li et al., 2026).

2.2.3. Algorithmic Information Cocoon Effects and Addiction Reinforcement

Algorithmic recommendations, by delivering highly personalized content, continuously amplify users' immersion and dependency (Wang & Han, 2025). Information cocoon effects restrict exposure to diverse information, thereby exacerbating cognitive narrowing and behavioral dependence, thus forming a self-reinforcing loop of "recommendation → immersion → increasingly precise recommendation → deeper immersion." This loop possesses inherent self-reinforcing characteristics: algorithms perpetually refine content delivery based on user behavioral data, precisely matching users' emotional needs and interest preferences, which significantly diminishes the motivation to disengage. At the social level, the social attributes of short videos (sharing, commenting, duet features) further reinforce usage stickiness through peer pressure, compelling users to persist

despite awareness of adverse consequences.

2.3. Impacts of Short-Video Addiction on College Students

The negative ramifications of short-video addiction encompass multiple domains, including physical and mental health, academic and cognitive performance, and value formation.

2.3.1. Impaired Sleep Quality

Research indicates a positive correlation between short-video addiction and insomnia. Nearly 60% of users engage with short videos before bedtime for prolonged periods; concentrated attention on video content maintains cerebral excitement, impedes sleep onset, significantly curtails sleep duration, and the impact intensifies with the duration of addiction (Lu & Yan, 2025).

2.3.2. Compromised Physical and Mental Health

Excessive short-video use is associated with unhealthy dietary patterns, elevated stress levels, and increased depression and anxiety. Prolonged immersion in fragmented information maintains the brain in a sustained state of hyperarousal, inflicting psychological harm. Users engaging in upward social comparison with others' meticulously curated "highlight reels" are prone to self-doubt and anxiety, yet remain unable to cease usage, fostering a paradoxical psychological state of "scrolling while simultaneously self-reproaching."

2.3.3. Declined Academic and Work Efficiency

Chronic short-video overuse impairs time management capabilities, encroaching upon study and work hours, impeding the timely completion of academic tasks, and diminishing academic performance. Excessive social media use distracts users from primary objectives, thereby adversely affecting scholastic achievement.

2.3.4. Cognitive and Value Impacts

Algorithmically constructed "information cocoons" contribute to user de-individuation, promoting passive information reception and attenuating active critical thinking (Wang & Han, 2025). Short-video addiction disrupts college students' study and life rhythms, potentially inducing negative emotional states and adversely affecting cognition, emotion, and value systems. Critics characterize short videos as "time sinkholes" that ensnare users and impede engagement with serious pursuits.

3. Multi-Dimensional Tiered Intervention Strategies

Based on the foregoing analysis of psychological mechanisms, this paper proposes a synergistic four-tier intervention framework encompassing individual, platform, institutional, and societal levels.

3.1. Individual Level: Cognitive Reappraisal and Systematic Self-Control Training

First, facilitate a shift in college students' temporal orientation from past/present focus toward future orientation. Evidence indicates that strategies aligned with future goals more effectively prevent addictive behaviors. Interventions may include future-oriented goal-setting training and time management planning, thereby activating initiatory self-control systems as an alternative to passive inhibitory strategies.

Second, identify and differentiate populations at risk for the "backfire effect." For college students with low inhibitory control, simplistic "usage prohibition" directives may prove counterproductive; priority should instead be accorded to alternative behavioral activation (e.g., sports participation, arranging real-world social activities) rather than direct self-restraint mandates.

Third, strengthen media literacy education. Assist college students in comprehending the underlying logic of algorithmic recommendations and "addictive design" strategies, enhancing their critical cognitive capacities regarding recommended content, and thereby restoring the primacy of rational decision-making.

3.2. Platform Level: Algorithmic Transparency and Anti-Addiction Design

Optimizing algorithmic recommendation systems addresses the technological root cause of addiction. Specific measures include: (1) introducing randomized content recommendations or user-customizable filters to break the closed structure of information cocoons; (2) embedding active exit nodes, such as intelligent usage duration reminders, timed automatic pauses, and enhanced natural exit signals, as alternatives to the current default "infinite scrolling" pathway; (3) enhancing recommendation algorithm transparency by providing users with explanations for content delivery (e.g., "Why this content?"), thereby transitioning algorithms from "black-box manipulation" to "visible assistance" (Zhao & Kou, 2026).

Article 25 of the EU DSA explicitly prohibits platforms from designing manipulative online interfaces that impair users' autonomous decision-making. China's *Administrative Provisions on Algorithmic Recommendation for Internet Information Services* similarly mandate that algorithmic recommendation service providers offer users options not based on their personal characteristics, as well as convenient functions to disable algorithmic recommendation services. Platforms should proactively align with international regulatory trends by incorporating user welfare into algorithmic objective functions.

3.3. Institutional Level: Mental Health Education and Campus Culture Development

Higher education institutions should construct systematic short-video addiction prevention frameworks: offering general education courses in media literacy to help students identify algorithmic manipulation and addiction signals; establishing short-video usage behavior screening mechanisms to identify high-risk groups and implement tiered interventions; and reinforcing mental health education to provide targeted counseling for antecedent variables of addiction, such as loneliness and anxiety. Concurrently, enriching campus cultural, sports, and offline social activities provides alternative channels for social support acquisition and value realization, fundamentally reducing emotional compensatory dependence on short videos.

3.4. Societal Level: Algorithmic Governance Regulations and Collaborative Intervention Networks

Short-video addiction results from the interaction of multiple systemic factors, necessitating robust policy regulation as a macro-level safeguard. Regulatory authorities should incorporate platforms' "addictive design" elements into algorithmic filing review scopes, establishing systematic risk assessment and mitigation obligations for algorithmic recommendations. Furthermore, it is imperative to promote the development of collaborative intervention networks involving families, educational institutions, communities, and platform enterprises: parents participating in the implementation of children's digital usage plans, communities providing offline activity spaces, institutions monitoring student behavioral changes and intervening opportunely, and platforms offering data support and substantively effective parental control tools for minors.

4. Conclusion

Short-video addiction among college students in algorithmic recommendation scenarios constitutes a complex phenomenon resulting from the interaction of technological design, individual psychological factors, and environmental influences. Its developmental mechanisms involve neural hijacking of the dopaminergic reward system, interactions among multiple psychological variables within the I-PACE framework, non-linear effects (e.g., the "backfire effect"), and self-reinforcing feedback loops of information cocoons. Effective intervention must transcend unidimensional "usage restriction" and shift toward multi-dimensional tiered strategies: activating initiatory self-control at the individual level, promoting anti-addiction design at the platform level, refining psychological education and literacy curricula at the institutional level, and improving algorithmic governance regulations at the societal level. Future research should further pursue longitudinal tracking to elucidate the dynamic trajectories of addiction development, explore adaptive intervention protocols based on individual heterogeneity, and construct a comprehensive prevention and treatment system encompassing "early screening—early warning—precision intervention."

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